

7.4.18

EE25BTECH11011 - Navya

Question:

If the lines $3x - 4y - 7 = 0$ and $2x - 3y - 5 = 0$ are two diameters of a circle of area 49π square units, find the equation of the circle.

1) $x^2 + y^2 + 2x - 2y - 47 = 0$

2) $x^2 + y^2 + 2x - 2y - 62 = 0$

3) $x^2 + y^2 - 2x + 2y - 62 = 0$

4) $x^2 + y^2 - 2x + 2y - 47 = 0$

Solution:

The general equation of a circle can be written as

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{b}^T \mathbf{x} + c = 0, \quad (1)$$

where

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \quad c \in \mathbb{R}. \quad (2)$$

The center $\mathbf{c}$ and radius r relate to $\mathbf{b}$ and c as:

$$\mathbf{c} = -\mathbf{b}, \quad (3)$$

$$r = \sqrt{\mathbf{b}^T \mathbf{b} - c}. \quad (4)$$

Since the lines are diameters, the center lies on both:

$$\mathbf{n}_1^T \mathbf{c} = d_1, \quad (5)$$

$$\mathbf{n}_2^T \mathbf{c} = d_2, \quad (6)$$

where

$$\mathbf{n}_1 = \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \quad d_1 = 7, \quad \mathbf{n}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad d_2 = 5. \quad (7)$$

In matrix form,

$$\begin{bmatrix} 3 & -4 \\ 2 & -3 \end{bmatrix} \mathbf{c} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}. \quad (8)$$

Solving for $\mathbf{c}$:

$$\mathbf{c} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (9)$$

The area of the circle is 49π , so radius is

$$r = 7 \quad (10)$$

and from the radius relation,

$$r^2 = \mathbf{c}^T \mathbf{c} - c, \quad (11)$$

$$49 = 1^2 + (-1)^2 - c = 2 - c, \quad (12)$$

$$\implies c = 2 - 49 = -47. \quad (13)$$

so the equation is

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{b}^T \mathbf{x} + c = 0, \quad (14)$$

which expands to

$$x^2 + y^2 + 2(-1)x + 2(1)y - 47 = 0, \quad (15)$$

Therefore, the equation of the circle is

$$\boxed{x^2 + y^2 - 2x + 2y - 47 = 0.} \quad (16)$$

